THE LOCALISATION MAP OF A RING AT A MULTIPLICATIVE SUBSET IS FORMALLY ETALE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The infinitesimal lifting formulation of formal etaleness. Unwrapped at the start of the counted proof (lines 41465-41468). Reduces the statement to a lifting condition for ring maps into a ring whose square-zero ideal the elements of S become invertible modulo.
- Units lift along a nilpotent thickening. Used at the end of the counted proof (lines 41468-41470). Shows that invertibility modulo a square-zero ideal implies invertibility, via the inverse image description of the units.

Lemma 0.1. *Let R be a ring. Let $S \subset R$ be any multiplicative subset. Then the ring map $R \to S^{-1}R$ is formally étale.*

Proof. Let $I \subset A$ be an ideal of square zero. What we are saying here is that given a ring map $\varphi : R \to A$ such that $\varphi(f) \mod I$ is invertible for all $f \in S$ we have also that $\varphi(f)$ is invertible in A for all $f \in S$. This is true because A^* is the inverse image of $(A/I)^*$ under the canonical map $A \to A/I$. □

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